Homework 1 Written Responses

Problem 4 – Infinite vs Finite Well Comparison

Why are the energy levels of the finite well lower than those of the infinite well?

In a finite well, the wave function is allowed to "leak" into the classically forbidden region (outside the well), making the effective width of the wave function larger than the physical width of the well. Because a particle's kinetic energy is inversely proportional to the square of its wavelength (or the effective width, $E \propto 1/L^2$), this spatial spreading decreases the kinetic energy. Consequently, the energy levels (relative to the bottom of the well) are lower than they are in an infinitely rigid box.

Why does tunneling occur in the finite well but not in the infinite well?

Tunneling is the phenomenon of a particle possessing a non-zero probability of being found in a region where its total energy is less than the potential energy. In an infinite well, the potential barrier is literally infinite, taking an infinite amount of energy to penetrate; hence, the probability immediately drops to strictly zero at the boundaries. In a finite well, the potential barrier is limited (10 eV), so the wave function decays exponentially but does not immediately vanish, allowing the particle to exist outside the well.

Compare the shapes of the wave functions near the boundaries.

For the infinite well, the wave functions have a sharp corner at the boundaries where they abruptly go to zero, forming a definite node. For the finite well, the wave functions are smooth and continuous at the boundaries, seamlessly transitioning from an oscillating sine/cosine wave inside the well to an exponentially decaying tail outside the well.

Which system more closely resembles real physical systems? Explain.

The finite well. In nature, no potential barrier is truly infinite. It always takes a finite (though sometimes very large) amount of energy to free a bound particle or overcome a barrier. For instance, an electron bound to a proton in a hydrogen atom has a finite ionization energy (13.6 eV), meaning it can escape if given enough energy, which matches the behavior of a finite well.

Problem 5 – Reflection

Physically, quantization implies that a confined particle cannot possess just any arbitrary amount of energy; rather, it is restricted to specific, discrete energy levels much like the distinct resonant frequencies of a plucked guitar string. This wave-like behavior directly arises from the application of boundary conditions. Boundary conditions—such as the requirement that a wave function goes to zero at an infinite barrier or remains continuous across a finite boundary—force the physical solutions of the Schrödinger equation to form standing waves. Only specific wavelengths (and therefore specific momenta and energies) can perfectly "fit" these mathematical constraints, demonstrating that physical confinement intrinsically generates energy quantization.